

Moon Sketching guide

Introduction

This guide was created because a lot of people use the telescopes to look at the Moon. And I don't blame them – it looks great! Plus, humans have been observing, tracking and gazing at the Moon for tens of thousands of years. However, many people will look at it, say “it looks cool”, and then move on. Instead, I want this guide to help people develop a long-term interest in the Moon, and to learn about its turbulent past, all just to show that a lot can be learnt about the Moon through amateur astronomy (no Apollo or Artemis spacecraft needed).

This guide will try to teach the reader to make sketches of the Moon's features by paying close attention to what they observe through a telescope. It'll try to teach you how to use varying light levels to study and judge crater depth, and it will also give you some craters and features to get started with. Sketching is a really fun way to document your observing and getting to know the Moon, and I hope this guide shows just that.

If you have any feedback on this guide, send it to the CUAS observation secretaries, and we'll use it to improve this and future guides 😊

P.S Don't forget the lunar filter!

Lan

CUAS Observation Secretary

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Map of the Moon

This is a map of what you'd see through the Dobsonian (which swaps north and south)
Not all the targets mentioned on this guide are labelled here....

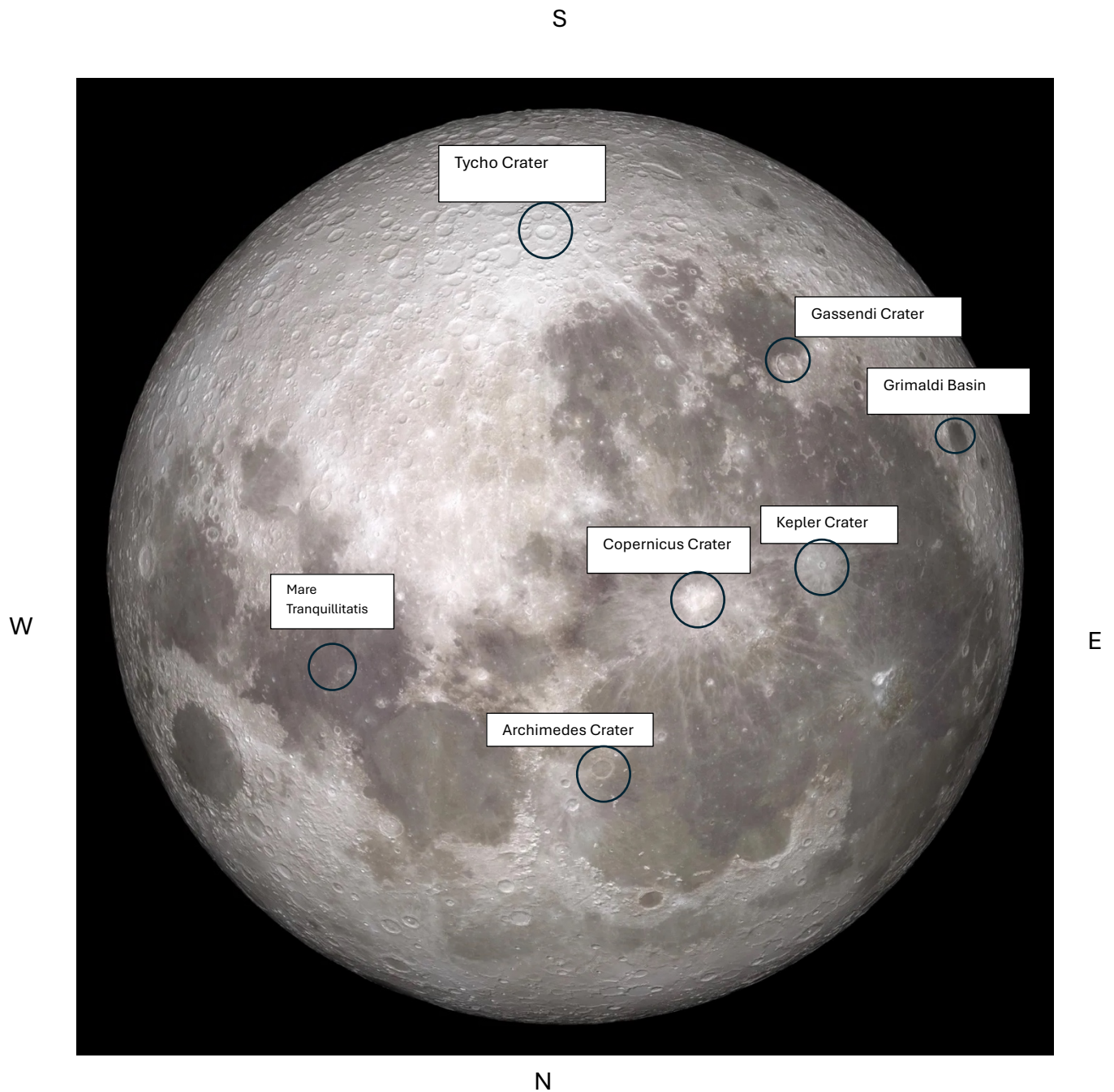


Image credit: Britannica

Who cares?

Why do we care about the Moon?

This question is quite subjective, and since humans have been observing it for tens of thousands of years, there's a wide variety of answers.

In ancient times, the Moon represented powerful deities:

- In Ancient Greece, the Moon represented the goddess Selene who was often depicted driving a silver chariot across the night sky.
- In Egyptian Mythology, the Moon was associated with the god Khonsu who controlled the moon's movement across the night sky and represented the passage of time, healing and protection
- In Chinese Mythology, the most famous lunar god is Chang'e, a symbol of beauty, immortality and longing.

Cut to the 1500s, and we have astronomers such as Galileo and Kepler studying the moon. Galileo discovered the Moon's surface wasn't in fact smooth, but instead filled with craters, and made a number of sketches too, which he included in his book *Sidereus Nuncius*, "The Starry Messenger".

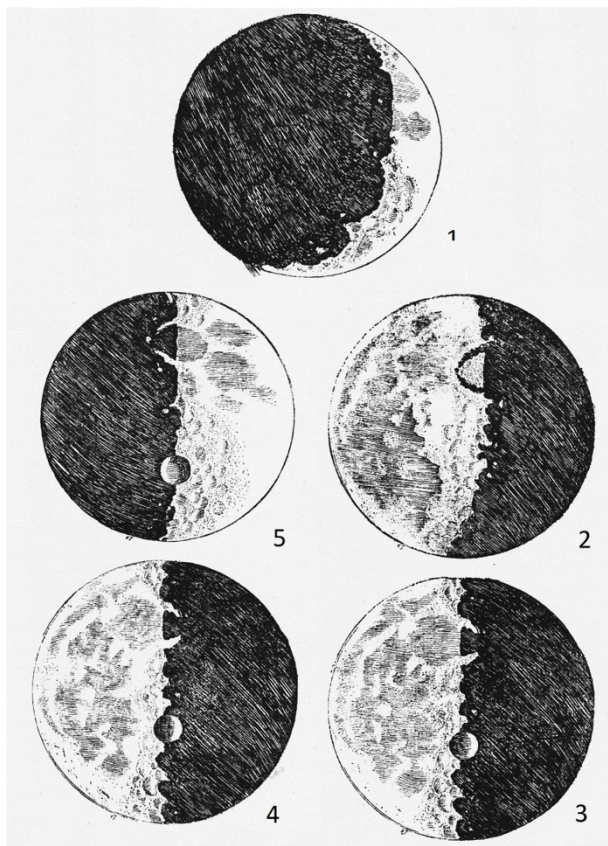


Figure 1: Drawings made by Galileo in his book *Sidereus Nuncius*

Kepler felt inspired by the Moon to write one of the earliest works of science fiction, titled *Somnium*. The novel is a tale about a journey to the moon. In the tale, the trip from Earth to the moon is believed to take around 4 hours. Extreme cold and lack of air are both concerns during the trip, so those making the journey are sedated and damp sponges put on their noses to allow them to breathe.

Kepler also speculated that the Moon caused tides on Earth, noting that they both followed similar 19-year cycles.

Then cut to the 1960s, and the beginning of the space race. The moon became a political symbol of strength and brain power, and the legacy/ impact of the first ever moon landing was summed up

nicely by Neil Armstrong as he stepped onto the lunar surface.

Today, space missions are returning to the Moon. The most recent Artemis mission caught a glimpse of the far side of the Moon, the end goal of the programme being to establish a long-term base on the lunar surface.

So fair enough, the Moon is interesting in many different ways for many different reasons. But why should we draw it? Why draw craters when we can photograph them anyway?

In my opinion, sketching the Moon is one of the best ways to truly understand the lunar surface for yourself. It's also one of the easiest things you can do to study the lunar surface – you don't need to spend money on astrophotography equipment or anything state-of-art, just a pencil, some paper, and a descent ground telescope will do. Physically drawing the features of the Moon gives you a better appreciation of the surface than just looking at photos.

Don't underestimate your eyes too. The Artemis II crew made observations of the lunar surface mainly using their eyes, because as the lead scientist for the mission Kelsey Young put it, "The human eye is basically the best camera that could ever or will ever exist".

I guess this is just a long way of saying that if you want to really appreciate the Moon's features, sketching is one of the best and easiest ways to develop familiarity with the lunar surface.

Getting Started

Equipment

A solid equipment list looks like this:

- **A pencil** - HB works fine, and you can get quite a large range of shades out of a HB. If you want your dark patches to look really dark, maybe use a B pencil. For outlines of different regions though HB is perfect.
- **Something to draw on** - You don't have to buy premium paper or any serious artist's equipment to make good sketches of the moon. A clip board and some A4 copier paper will do (but it's probably best to use paper that doesn't wrinkle too much due to the moisture in the air). Equally, notebooks are good (preferably plain paper ones just so your sketches are clearer). You can also make sketches in the Northumberland observation book.
- **Something to look at the moon in detail with** – Ideally, you'll have your hands free. The Northumberland is a good choice because it will track the moon for you, the Dobsonian is easier to set up but has the slight disadvantage of having to adjust it every so often. Binoculars on a tripod might not give you enough magnification, although they are the simplest to use. It's definitely worth experimenting with what you have and seeing what gives you the best view of what you want to observe. You don't have to have the world's best telescope, just a lot of ambition!

A couple more handy tips:

Decide what you want to observe in advance - The whole process is a lot more efficient if you already know what you want to look at. This way you can set up your telescope, find what you're looking for, and devote 90% of your observing time to making a detailed and accurate sketch.

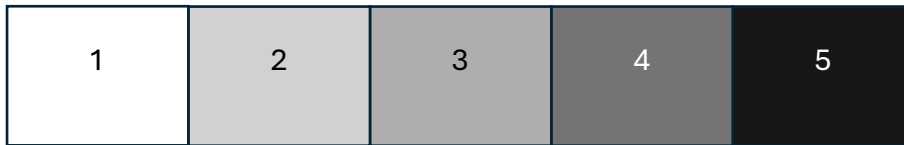
Another very important point is to **never forget to use a lunar filter**, especially if you're using the Dobsonian or the Northumberland. The sun light reflecting off the Moon is bright and won't do your eye's much good if you stare at it too long, so protect your vision and use a filter.

Also, making sketches in the dark is difficult, but you also don't want to be sketching in bright light that will ruin your night vision. A small source of red light is an ideal compromise.

Making a Sketch

After you've decided your target, choose a greyscale that lets you see the intensity of light on the lunar surface. In the beginning, a 5-point scale is probably easiest to use, just so you can get your eye in noticing obvious differences in light and dark. The main thing to focus on when starting out is just accurately drawing the crater shape. Before long you'll be able to add more and more shades of grey to your scale as you become better at noticing details (maybe don't aim for more shades than 10 though, otherwise it's easier to lose the definition of the features you're trying to draw)

An example of a scale to use:



Drawing this in the corner of your page before you begin helps you decide which region has what shade.

The numbers are useful too – when you make your sketch, you'll probably do the actual shading when you get home. When you're at the telescope, it's better to just number the different regions you've outlined according to how light or dark they are, so that you're left with a colour-by-numbers drawing to fill in later. Again, it's important to make sure your outlines are accurate, just so that your sketch resembles what you're actually looking at.

All of this is probably best illustrated with an example. I am going to cheat slightly and draw from a photo of the moon taken with my phone whilst using the Dobsonian, but all of the steps I take here should be applicable to just making sketches directly from a telescope. I do take advantage of the fact that you can zoom in on a photo, but this is equivalent to using higher magnification, so you should be able to get similar results.

The photo I'm going to use is below. A few notable features:

- The **terminator line** is the line separating the Moon's day side from its night side. As is clear from the photo, it's the best place to find really defined craters. The light from the sun is at just the right angle here to let the shadows reveal the crater depth. This shows that trying to sketch features during a full moon is difficult because everything looks bright (compare the terminator line with the Tycho crater). So, once you have a crater/ target in mind, try and time your observing so that the terminator line falls across it. It's also a nice idea to draw the same target at different light levels as this will also give you better feel for the target's 3D shape.

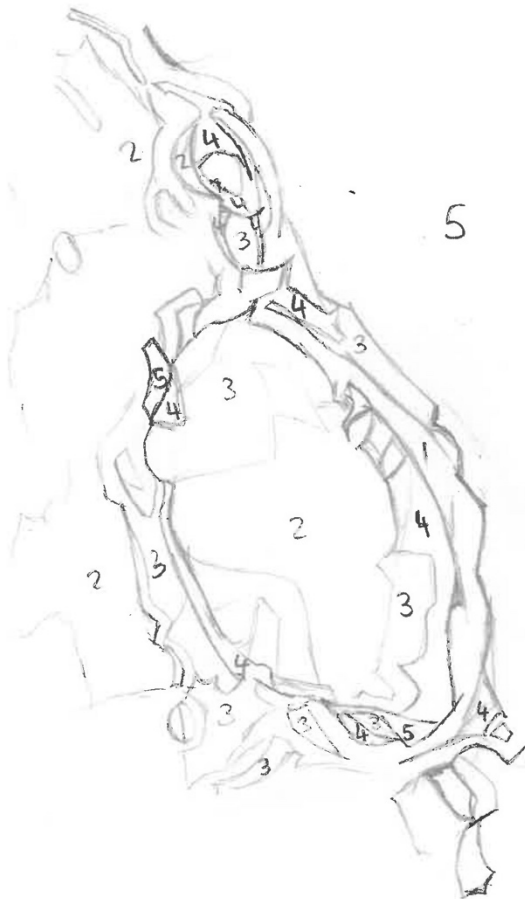


Figure 2: The Moon! (taken using the Dobsonian and my phone)

- The Tycho crater is one of the most recognisable features, mainly because of the bright streaks/ rays coming out of it. Other notable features are the Copernicus and Kepler craters, but more about them later....

Zooming in on the crater in the red circle:





To do at the telescope:

My first step is to draw an outline of the crater. Take your time over this bit - you'll probably have to have a few attempts before you get it just right. Then you want to add your grey scale numbers – remember to do every little region. As a result, it's a good idea to draw fairly big.

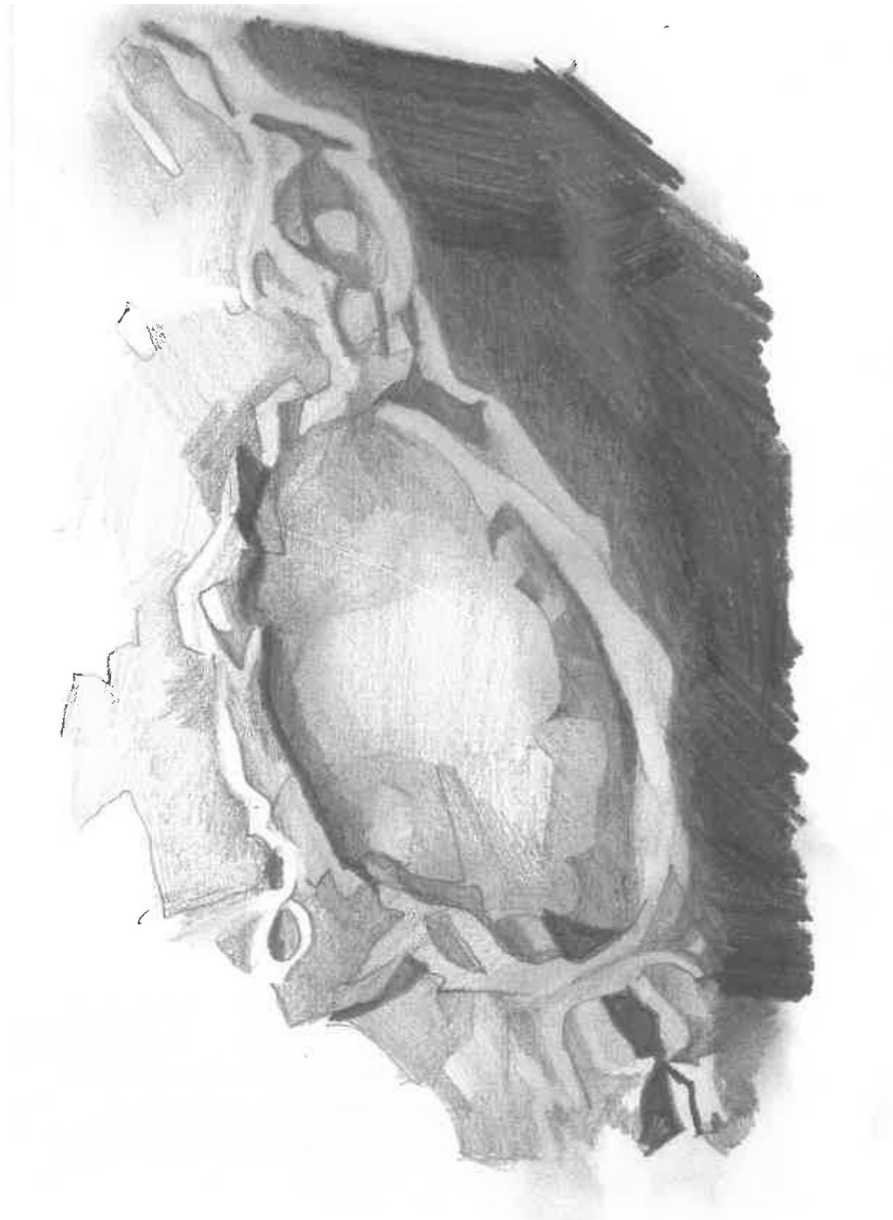
You should also do what I forgot to do here and write down the name of the crater you're drawing, the date and time you observed it, the telescope you used and the magnification.

Once you've got this down, you can move onto your next target, go home and eat, sleep, etc.



To do at home:

Trace out your previous line drawing so that you're just left with the outlines of each of the different regions. This is what you'll shade in eventually, instead of the one with numbers. The one with numbers is your guide.



Then you can finally shade in your drawing. In hindsight I probably would have used a darker shade for the dark side of the terminator line, just to define the edges a little more... I would have also smoothened out the shading of the inside of the crater. Practice makes perfect.

If you like you can also add a little scale next to your drawing. In reality the size of the drawing I did was around A6 size, but experiment with whatever size drawings suit you best and allow you to note enough detail the easiest.

Things to see

Now we move onto a list of interesting Moon features to sketch. A note on the location maps (drawn in pencil) – the black circles indicate the general region of the feature, and don't necessarily show the feature's size. The Moon also wobbles a little as it goes through the Lunar phases, partly due to its slightly elliptical orbit, and partly due to the Earth's own rotation shifting the position from which we view the Moon. So, the black circles may also be a little wide/ big too.

The maps also show the view from the Dobsonian which flips North and South – it's the same thing that you'd see if you were looking at the Moon from the southern hemisphere. Also, T marks the position of the Tycho crater, C the Copernicus crater and K the Kepler crater.

If I haven't included a map, then refer back to the image where I point out the terminator line.

Craters

Tycho Crater

Tycho is one of the most prominent craters on the moon, not because of its size but because of its age – the bright rays/ streaks from Tycho show that it is relatively young. It's believed that some of the moon rock samples taken on Apollo 17 were from Tycho, and they were found to be around 108 million years old, significantly younger than the 3.9 billion years for many other moon craters.

The crater is sharply defined and well preserved. The small peak in the middle of the crater is thought to have been formed from material that rebounded after impact. Tycho cools more slowly than other parts of the moon, due to the materials that cover the crater, which includes aluminium-rich anorthosite.

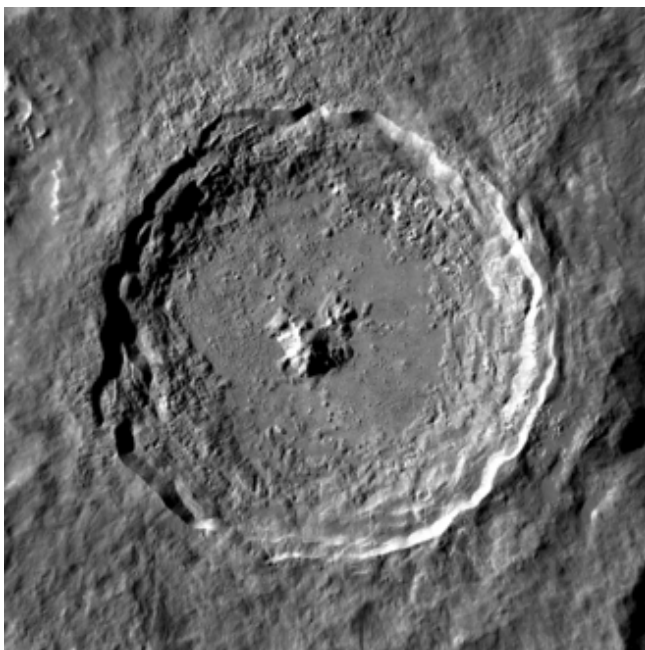


Figure 3: Tycho as seen by the Lunar Reconnaissance Orbiter



Figure 4: Tycho from my photo

Plato Crater

The smooth flat floor of Plato is due to lava flooding the crater many years ago. Sunlight passing over the crater produces elongated shadows near its eastern rim, and the crater is around 3.8 billion years old. The base of this crater rivals the Grimaldi basin for the darkest spot on the moon, and in the past astronomers believed Plato's crater to be a sea.

Keep a look out for Plato's Hook, a mysterious hook shaped shadow pattern that appears on the floor of the crater under certain light conditions. The shadow is cast by a 3km high peak known as the Gamma Peak on the eastern rim of the crater.

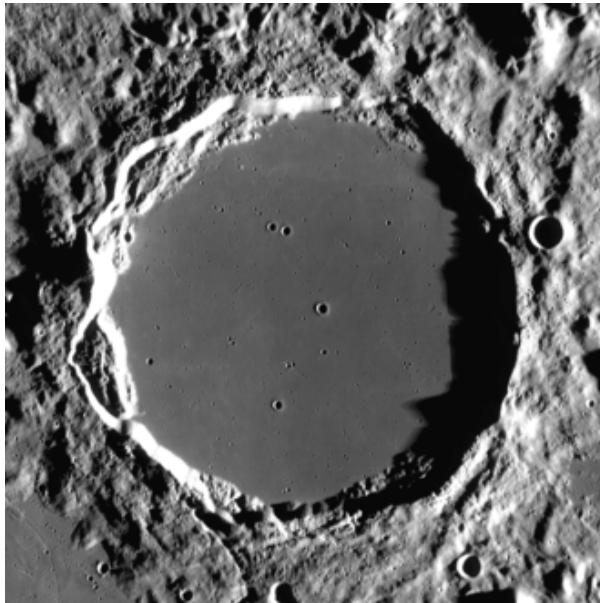
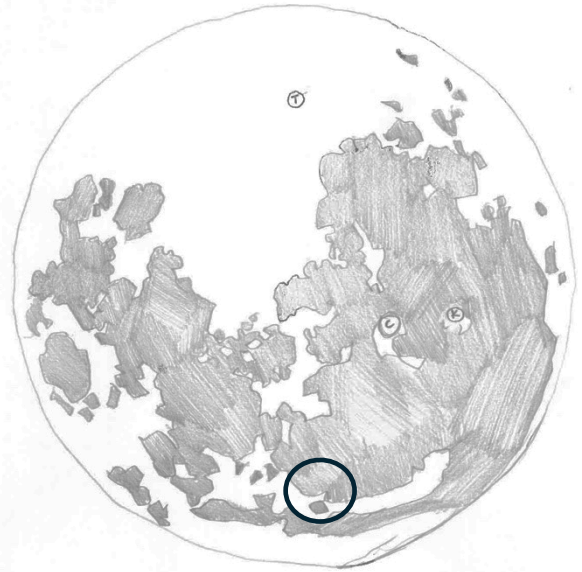


Figure 5: Plato Crater as seen by the Lunar Reconnaissance Orbiter



Located near the base of the moon

Copernicus Crater

This crater is also known for its bright rays and is around 90km across. Less than 1 billion years old, it has no central mountain/ peak but there are many hills near the middle of the floor. The rays from Copernicus extend to at least 805km, and the crater remains bright as it is relatively young and has not yet been darkened by meteorite bombardment or solar wind effects.

Compare the rays of Copernicus with those of Tycho, and you'll notice that rays around Copernicus are more irregular. The crater has the nickname "Monarch of the Moon". Many of the Moon's craters were named by Giovanni Riccioli, an Italian astronomer from the 1600s. He named the crater after Copernicus after Copernicus suggested his heliocentric mode of the solar system, which Riccioli disagreed with on religious grounds. Riccioli justified the name by showing that he had thrown all heliocentrist astronomers into the Ocean of Storms, but some historians also believe that this was Riccioli's way of ensuring that Copernicus' legacy was remembered.

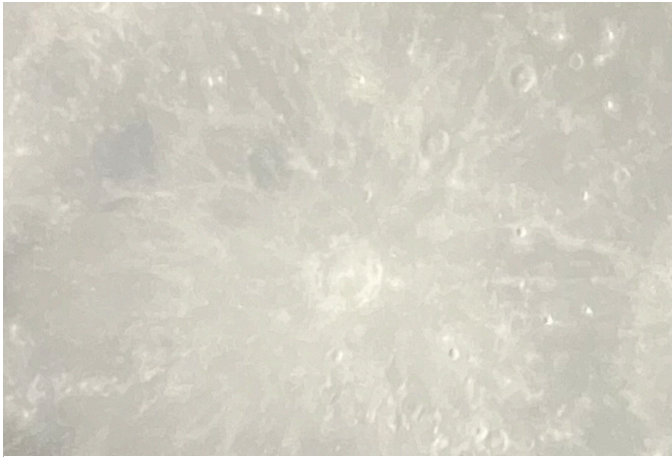


Figure 6: Copernicus Crater (central)

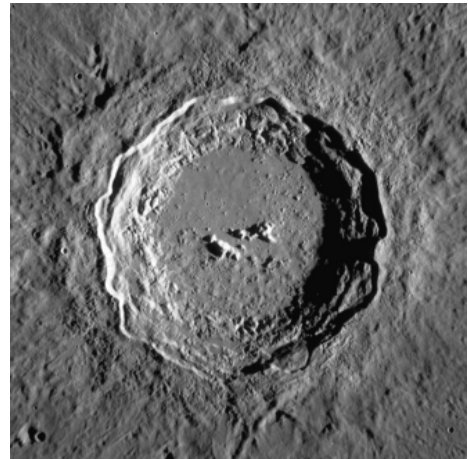


Figure 7: Copernicus Crater as seen by the Lunar Reconnaissance Orbiter

Kepler Crater

Kepler's rays extend well over 300km, and the crater itself has an uneven floor. One of Tycho's rays intersects the crater, and this led to the crater's name, which is symbolic of Kepler's usage of Tycho Brahe's astronomical data when developing his three laws of planetary motion. The crater is 32km across and 2.7km deep.

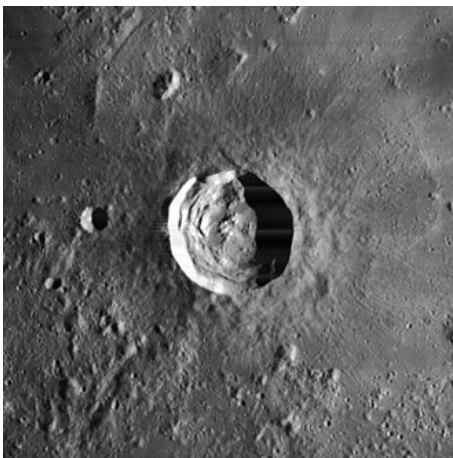


Figure 8: Kepler Crater as seen by Lunar Orbiter 4



Figure 9: Kepler next to Copernicus

Grimaldi's Floor

Said to be the darkest patch of the Moon, this crater is 3.9 billion years old and 222km across. The crater hides a "mascon" - a mass concentration or region beneath the surface which has a higher density than other regions of the moon. The inner wall has

been reduced over time to irregular hills, ridges and peaks, although some peaks range up to 2km in height.

Grimaldi's floor is known for transient behaviour, including gas emission, flashes of light and the appearance of colour patches.

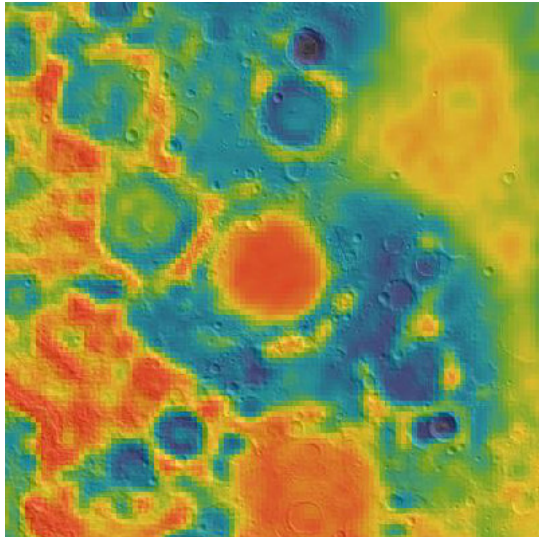


Figure 10: a gravity map of Grimaldi's Floor (red. = high, blue = low) (see references for credit)



Figure 11: Grimaldi's crater as seen during the Artemis II mission (credit NASA)



Look out for the darkest patch of the moon – it'll be Grimaldi's basin.

Archimedes Crater

83km in size and over 3 billion years old, the walls of this crater rise to just over 2km above the crater floor. The crater floor is smooth like the Plato crater, as both flooded with lava at some point, but Archimedes is much lighter in colour. The crater has a circular shape but appears elliptical from Earth due to foreshortening.

The first man-made object to land on the moon lies near the Archimedes Crater: in 1959, the Soviet-launched Luna 2 crash landed onto the Moon's surface, although exactly where the spacecraft landed we're not so sure, as it never sent back a signal.

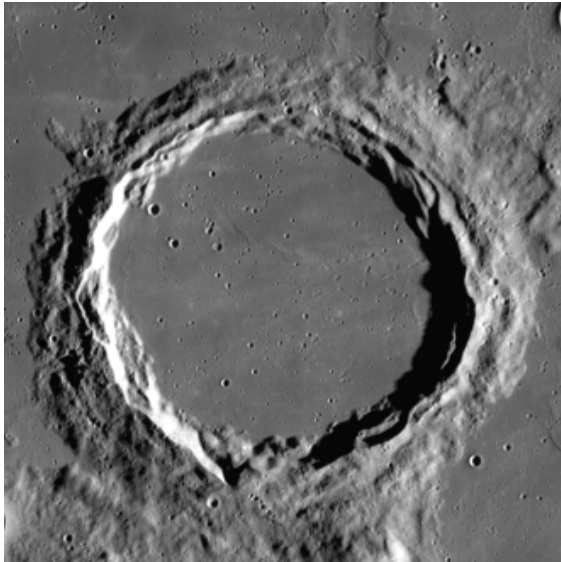


Figure 12: Archimedes as seen by the Lunar Reconnaissance Orbiter



Located southwest of Copernicus

Others

Alpine Valley

A straight valley 188km long with a flat floor through the Moon's Alps mountain range. It was once thought to have been the result of a very low-angle asteroid impact, but now it is believed to be an area where the Moon's crust dropped downwards between 2 parallel fault lines known as a graben. The centre channel is hard to find using a telescope, however.



Figure 13: Vallis Alpes as seen by Lunar Orbiter 4



Located beneath the Archimedes Crater

Apennine Mountains

Probably the most stunning of the Moon's mountain ranges, the Apennine Mountains span a curve that's around 600km long. Its peaks rise to heights of around 5.5 km and face the flat plains of the Mare Imbrium. The range makes up some of the southeastern part of the rim of the Imbrium Basin. The Mountain range is around 3.8 billion years old.

Nearby, another geological feature bordering the Imbrium Basin is the Bay of Rainbows, or *Sinus Iridium*. It's thought to have once been a crater, but one wall was flooded with

lava leaving a bay. You can watch as the sun rises over this feature and creates what's known as the Jewelled Handle.

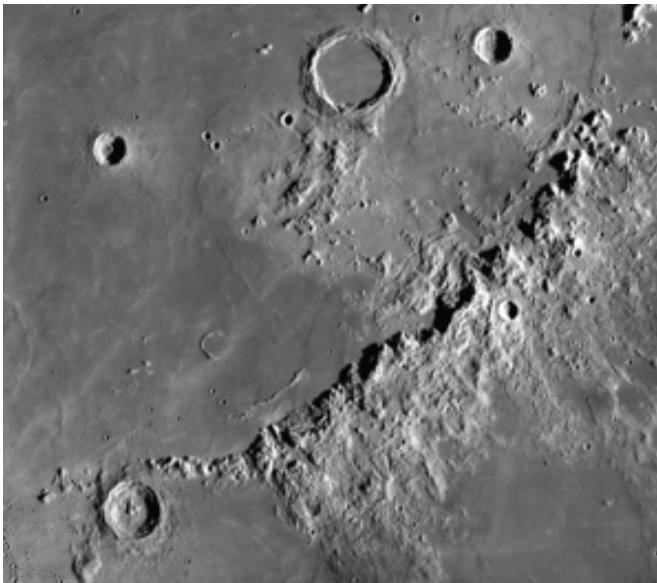
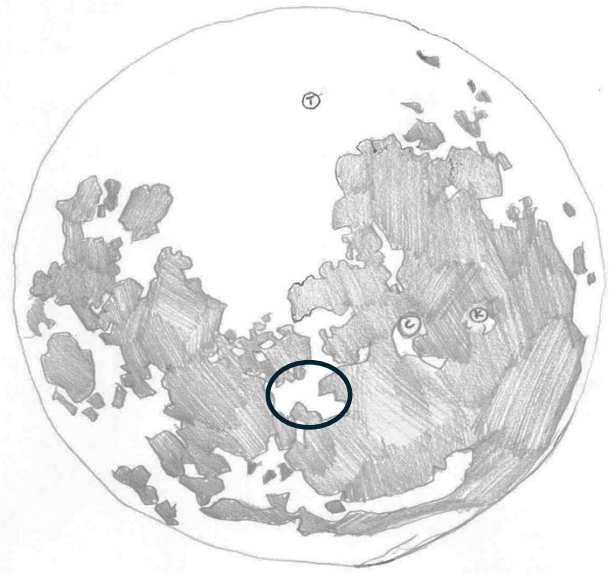


Figure 14: Apennine Mountains as seen by the Lunar Reconnaissance Orbiter



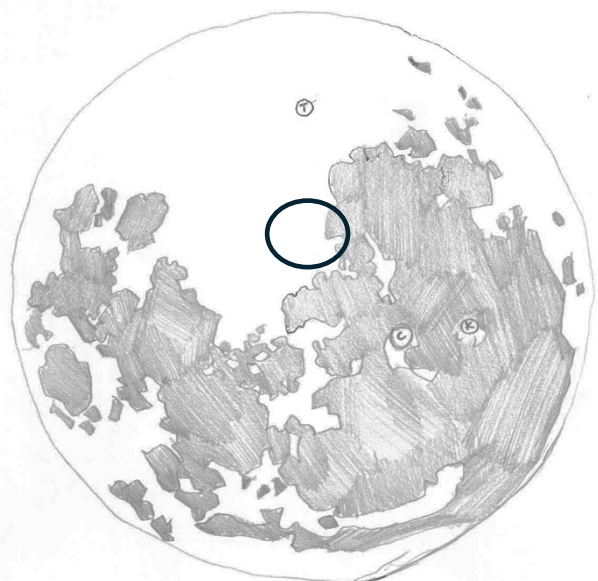
Located above the Archimedes crater

Catena Davy

Catena Davy is a long chain of 20 craters spanning a distance of 45km. The craters themselves are up to 2km wide and thought to have been made when an asteroid broke up in the Moon's / Earth's gravity, before slamming down onto the moon's surface. Volcanic activity can also cause such chains, but the Davy craters don't appear to be volcanic. Ejecta from asteroids can also cause such chains, but the Davy craters don't appear to be near any major impacts.



Figure15: Catena Davy as seen by the Lunar Reconnaissance Orbiter



Located beneath the Straight Wall

The Straight Wall

Also known as Rupes Recta, the Straight Wall is (as its name suggests) a very linear feature on the surface of the moon. The line becomes obvious under the right lighting. After a quarter of the Moon has been illuminated, the wall appears like a dark line. As the sun illuminates more of the Moon, the line becomes brighter and brighter. This is because the wall is actually a slope, around 110km long and 2-3km wide. The Wall marks a fault line on the Moon, a place where one tectonic plate has fallen below another

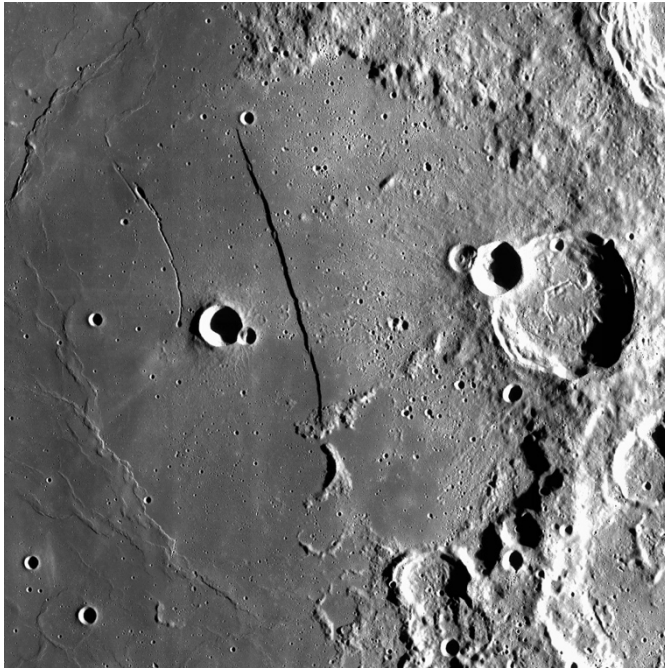
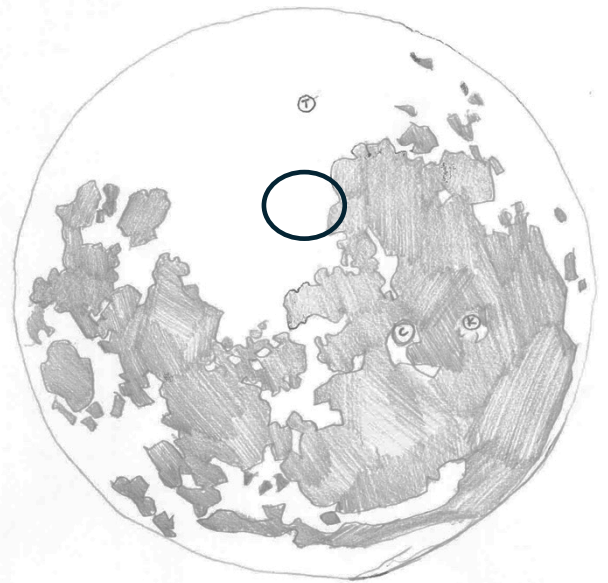


Figure16: Rupes Recta (the Straight Wall) as seen by the Lunar Reconnaissance Orbiter



Located beneath Tycho

References/ cool links

- Lunar deities: https://en.wikipedia.org/wiki/Lunar_deity
- Figure 10: By James Stuby / NASA / Arizona State Univ. - Transferred from en.wikipedia to Commons. (Original text: JMARS), Public Domain, <https://commons.wikimedia.org/w/index.php?curid=47554490>
- A really useful guide to features of the moon is NASA's Daily Moon guide: <https://science.nasa.gov/moon/daily-moon-guide/#1782807851819::0::> - It gives you daily recommendations of what to look at, and there's different options for telescopes, binoculars and naked eyes, as well as northern/ southern hemisphere options. It's really good to use when you're getting used to what's where on the Moon's surface.
- A lot of the info about craters and different moon features was taken from BBC Sky at Night Magazine: <https://www.skyatnightmagazine.com> - Patrick Moore's articles are always a good read, in addition to Pete Lawrence's. FYI Patrick Moore mapped the moon by hand, and those maps were used by NASA to prepare for lunar landings in the Apollo missions. You can read more about that here: <https://www.skyatnightmagazine.com/space-science/patrick-moore-mapping-the-moon> . He also played the Xylophone: <https://www.youtube.com/watch?v=WVGwJRvpTyQ> . I welcome any Pete Lawrence facts that can be added here ☺
- As already mentioned, NASA have great resources on learning more about the moon, as well as great photos from the Lunar Reconnaissance Orbiter and Lunar Orbiter 4, to name a couple. They also have their own guides on what to look at: https://assets.science.nasa.gov/content/dam/science/psd/earths-moon/observe-the-moon/Moon_Map_2026_Southern.pdf?emrc=6a313bbe9b35c
- NASA's page for all things lunar: <https://science.nasa.gov/moon/>
- The British Astronomical Association (BAA) have a shortish guide on sketching the Moon also: https://britastro.org/section_information_/lunar-section-overview/lunar-section-tutorials/the-art-of-drawing-the-moon
- Several Wikipedia articles on famous astronomers were used:
 - Copernicus: https://en.wikipedia.org/wiki/Nicolaus_Copernicus
 - Kepler: https://en.wikipedia.org/wiki/Johannes_Kepler
 - Kepler's novel Somnium: [https://en.wikipedia.org/wiki/Somnium_\(novel\)](https://en.wikipedia.org/wiki/Somnium_(novel))
- Also, read this article by the BAA about Galileo sketching the moon: https://britastro.org/journal_contents_ite/a-chronology-of-some-lunar-drawings-by-galileo-galilei
- Info about NASA's Artemis missions can be found here: <https://www.nasa.gov/humans-in-space/artemis/>